

Truong Giang Nguyen

ML Engineer — Production ML Systems & Model Deployment

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Family Visa (valid until March 2028) • No sponsorship required • Immediately available

SUMMARY

ML Engineer with 4+ years building and deploying production ML systems across computer vision, NLP, and predictive analytics. Deployed 2 production systems (price intelligence platform, analytics platform) using GCP infrastructure (Vertex AI, BigQuery). Strong model development capabilities combined with deployment experience using FastAPI, Docker, and ML pipelines (Airflow). Versatile across ML domains: computer vision (object detection), NLP (RAG, LLM fine-tuning), time-series forecasting, and predictive modeling. MSc in Data Science (Distinction) with proven ability to take models from experimentation to production deployment.

Core Expertise: Production ML Deployment • GCP (Vertex AI, BigQuery) • ML Pipelines (Airflow) • Model Development • Computer Vision • NLP & LLMs • FastAPI • Docker • CI/CD • Python/SQL • Model Monitoring

WORK EXPERIENCE

ML Engineer / Data Scientist

Seedcom R&D

Nov 2020 – Oct 2024

Vietnam (Remote)

Production ML Systems (Deployed):

- **Led team of 3 engineers** developing automated price intelligence system with multimodal AI; built product matching using Google SigCLIP (text + image embeddings with category-specific weighting) to match 6,000 SKUs across 6 retailers; **deployed to production** on GCP (Vertex AI), running 2+ years; integrated with BigQuery for real-time analytics dashboards.
- Contributed as team member (7-person team) to large-scale analytics platform **deployed to production** for Vietnam's largest supermarket chain (Kingfood/Onelife); optimized BigQuery pipelines handling millions of daily transactions; built production dashboards with FastAPI backend.

ML Infrastructure & Pipelines:

- Designed and automated ML pipelines on Google Vertex AI with orchestration using Airflow; implemented CI/CD practices with GitLab CI for automated model training and deployment; deployed FastAPI-based ML services handling inference requests; containerized models with Docker for reproducible deployments.
- Built monitoring infrastructure tracking model performance; documented deployment procedures and best practices; maintained production systems ensuring 2+ years of reliable operation.

Model Development (Computer Vision & NLP):

- Led team of 3 engineers developing fashion detection system (RetinaNet, 92% AP, 19 categories) for 2 years; evaluated YOLO/ResNet before selecting RetinaNet for production; implemented complete CV pipeline from data preprocessing to inference.
- Designed NLP systems: Vietnamese recipe chatbot using RAG with 800 recipes; personalized notification system using LoRA fine-tuning on Llama 3; demand forecasting models (DeepAR, LightGBM); demonstrated versatility across ML domains.

Independent AI/ML Consultant

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Oct 2024 – Present

- *Family Visa (no sponsorship required). Immediately available for full-time work.*
- Created multi-agent AI system using Google ADK with 100% routing accuracy; implemented safety guardrails and LLM-as-a-Judge evaluation framework; published on GitHub (8 stars) and YouTube tutorials.
- Maintained production-ready skillset: survival analysis (Cox PH, 0.6688 C-index on 28,800 patients), customer segmentation (Random Forest, 94.6% accuracy), LLM fine-tuning (LoRA on clinical notes).

TECHNICAL SKILLS

Cloud & ML Infrastructure: Google Cloud Platform (Vertex AI, BigQuery, Cloud Storage) • Production ML Deployment • Model Serving & Monitoring • Data Pipeline Engineering • ETL/ELT Processes • Cloud Cost Optimization

ML Ops & Deployment: ML Pipelines (Airflow orchestration) • CI/CD (GitLab CI) • Docker Containerization • FastAPI Services • Model Versioning • A/B Testing Infrastructure • Model Performance Monitoring • Data Drift Detection

Model Development: Computer Vision (CNNs, RetinaNet, Object Detection) • NLP & LLMs (RAG, Fine-tuning with LoRA) • Time-Series Forecasting (DeepAR, LightGBM) • Classification & Regression • Clustering • Ensemble Methods • Model Optimization & Quantization

Programming & Data Engineering: Python (pandas, NumPy, scikit-learn, PyTorch, TensorFlow) • SQL (query optimization, complex joins) • Data Preprocessing • Feature Engineering • API Development (FastAPI) • Data Visualization • Git Version Control

Production Systems Experience: 2 Deployed Production Systems • Real-time Inference Services • Dashboard Development • Database Optimization • System Monitoring • Documentation & Best Practices

KEY PROJECTS

Production-Ready ML Pipeline for Customer Segmentation (Jan 2025 - Mar 2025) — *Portfolio Project*

- Engineered end-to-end production-ready ML pipeline with automated data preprocessing, feature engineering, and model training; deployed Random Forest classifier achieving 94.6% accuracy and K-Means clustering for behavioral segmentation; built visualization dashboard with 20+ interactive charts; demonstrates full ML lifecycle from data ingestion to production deployment readiness.

LLM Fine-tuning with LoRA for Clinical NLP (Aug 2024) — *Model Optimization & Deployment*

- Fine-tuned Mistral 7B model using parameter-efficient LoRA on 4-bit quantization for clinical note summarization; improved ROUGE-1 from 0.381 to 0.423 (11% improvement); achieved 17.9-minute training time; optimized model for production deployment in resource-constrained environments; demonstrates expertise in modern LLM optimization techniques including quantization and efficient fine-tuning.

Multi-Agent AI System with Google Gemini (Oct 2024) — *Agentic AI Architecture*

- Designed and built demonstration multi-agent customer support system using Google Gemini with Master Agent orchestration coordinating specialized Policy Agent (RAG) and Ticket Agent (database operations); achieved 100% routing accuracy across 15+ test scenarios with sub-2-second response times; demonstrates expertise in modern agentic AI architectures and LLM orchestration patterns.

EDUCATION

Master of Science (MSc) in Data Science (Distinction) Jan 2022 – Jul 2023
Northumbria University, Newcastle upon Tyne, UK

Dissertation: "Assessing Effectiveness of Wearable and Smartphone Cues for Drooling in Parkinson's Disease"

Bachelor of Engineering (BEng) in Electrical & Electronic Engineering (2.1) Sep 2015 – Jun 2020
Ho Chi Minh City University of Technology, Vietnam

PUBLICATIONS

Nguyen-Truong, G., Le-Tien, T., Nguyen-Thanh, T., Phan, H.X., Ta-Quoc, V. (2020). "Deep Learning Based Approach Implemented to Image Super-Resolution." *Journal of Advances in Information Technology*, Vol. 11, No. 4.